



Practice Assignment 7

Instruction:

** Students are allowing to write their answers (like SQL queries, Screen shot of outputs, etc.) in word file (Answer sheet) provided by instructor. After finishing the assignment, students must convert the word file (Answer sheet) into a PDF file. Finally, students upload the file in Moodle.*

1. Writing Function Stored the following tables in a new database ‘Assignment3’:

Hàm đơn trị

```
CREATE FUNCTION name_function (Var)
RETURNS data_type
DETERMINISTIC
BEGIN
    Statement SQL
    RETURN value
END
```

1. Find the average salesman's salary.
2. Find the name of the highest paid salesman.
3. Find the name of the salesman who is paid the lowest salary.
4. Determine the total number of salespeople employed by the company.
5. Compute the total salary paid to the company's salesman.
6. Find Clients in a Province
7. Calculate Total Order Amount

2. Creating constraint for database “Assignment2”:

- a) Check constraint to value of gender in “Nam” or “Nu”.
- b) Check constraint to value of salary > 0.
- c) Check constraint to value of relationship in Relative table in “Vo chong”, “Con trai”, “Con gai”, “Me ruot”, “Cha ruot”.

3. Writing SQL Queries by database “Assignment2”

- a) Look for employees with salaries above 25,000 in room 4 or employees with salaries above 30,000 in room 5.
- b) Provide full names of employees in HCM city.

- c) Indicate the date of birth and address of Dinh Ba Tien staff.
- d) The names of the employees of Room 5 are involved in the "San pham X" project and this employee is directly managed by "Nguyen Thanh Tung".
- e) Find the names of department heads of each department.
- f) Find projectID, projectName, projectAddress, departmentID, departmentName, departmentID, dateOfEmployment.
- g) Find the names of female employees and their relatives.
- h) For all projects in "Hanoi", list the project code (projectID), the code of the project lead department (departmentID), the full name of the manager (lastName, middleName, firstName) as well as the address (Address) and date of birth (dateOfBirth) of the Employees.
- i) For each employee, include the employee's full name and the employee's line manager.
- j) For each employee, indicate the employee's full name and the full name of the head of the department in which the employee works.
- k) Provide the employee's full name (lastName, middleName, firstName) and the names of the projects in which the employee participated, if any.
- l) For each scheme, list the scheme name (projectName) and the total number of hours worked per week of all employees attending that scheme.
- m) For each department, list the name of the department (departmentName) and the average salary of the employees who work for that department.
- n) For departments with an average salary above 30,000, list the name of the department and the number of employees of that department.
- o) Indicate the list of schemes (projectID) that has: workers with them (lastName) as 'Dinh' or , whose head of department presides over the scheme with them (lastName) as 'Dinh'.
- p) List of employees (lastName, middleName, firstName) with more than 2 relatives.
- q) List of employees (lastName, middleName, firstName) without any relatives.
- r) List of department heads (lastName, middleName, firstName) with at least one relative.
- s) Find the surname (lastName) of unmarried department heads.

- t) Indicate the full name of the employee (lastName, middleName, firstName) whose salary is above the average salary of the "Research" department.
- u) Indicate the name of the department and the full name of the head of the department with the largest number of employees.
- v) Find the full names (lastName, middleName, firstName) and addresses (Address) of employees who work on a project in 'HCMC' but the department they belong to is not located in 'HCMC'.
- w) In general verse 16, find the names and addresses of employees who work on a scheme in a city but the department to which they belong is not located in that city.